Derek Lund

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EDUCATION

UCLA Samueli School of Engineering Expected June 2028

B.S. Aerospace Engineering

- Awards: Ann Marie Schladen Memorial Alumni Scholarship, Mu Alpha Theta Scholarship

Orange Coast College August 2024 — May 2026

Aerospace / Mechanical Engineering

- Activities: Mu Alpha Theta (Director of Activities), Coast Aeronautical/Astronautical Research Lab (Aerodynamics Team Lead)
- Relevant Coursework: Solid Fuel Propulsion, Differential Equations, MATLAB, Statics, Multivariable Calculus, Linear Algebra

SKILLS

Software & Programming: ANSYS Fluent, OpenRocket, OnShape, Siemens NX, MATLAB, Python, PowerShell, Arduino, Excel

Aerospace: CFD Aerodynamic Analysis, wind tunnel & flight data analysis, rocket structures, avionics, recovery systems

Methods: mathematical modeling, data analysis, scientific writing, LaTeX

EXPERIENCE

3D-printed TVC Mount — Independent Rocketry Project

June 2026 – Present

- Designed full OnShape CAD assembly of a 2-axis gimbal mount, avionics bay, and airframe using MATLAB for design constraints
- 3D-printed parts using ASA and a resin printer and used heat-set inserts to thread holes for fasteners
- Implementing Teensy 4.0 microcontroller, MPU6050 IMU, and servos and developed control software with PID tuning

Aerodynamics Team Lead — IREC Competition | Coast Aeronautical/Astronautical Research Lab

November 2025 – Present

- Conducted comparative CFD studies in ANSYS Fluent to evaluate aerodynamic performance across multiple rocket geometries
- Designed full-scale CAD models of selected configurations based on CFD-driven aerodynamic optimization results
- Built modular 3D-printed subscale models for wind tunnel validation of simulated force and stability predictions
- Analyzed aerodynamic forces and pitching moment trends across subsonic and transonic flight condition

L1 Rocket — Independent Rocketry Project

September 2025 – May 2026

- Built and flew a Level 1 high-power rocket, gaining hands-on experience with structural assembly and recovery systems
- Created a full CAD assembly and simulated flight trajectories in OpenRocket to optimize recovery deployment timing
- Applied aerodynamic stability principles and flight analysis methods to validate real-world vehicle performance

Independent Mathematics Research

June 2025 – Present

- Coauthored a mathematical proof involving triangle intersections, structured through four independent analytical lemmas
- Investigated graph theory applications related to transistor network layouts and data transfer optimization problems

IT Intern | Mainfreight Inc.

June 2025 – August 2025

- Automated migration workflows for more than 1,000 users from Work Folders to OneDrive using PowerShell
- Assisted with group policy administration and developed Excel-based tracking systems for project milestone management

Research Presenter | Giles T. Brown Symposium

September 2024 – May 2025

- Compared models of gravitational interactions in binary systems and generated MATLAB visualizations for technical presentation
- Presented analytical findings to faculty evaluators while strengthening technical communication and research skills

Independent Tutor

January 2024 – Present

- Provided individualized mathematics and science tutoring for middle school and high school students.
- Adapted instructional approaches for both online and in-person learning environments based on student needs.